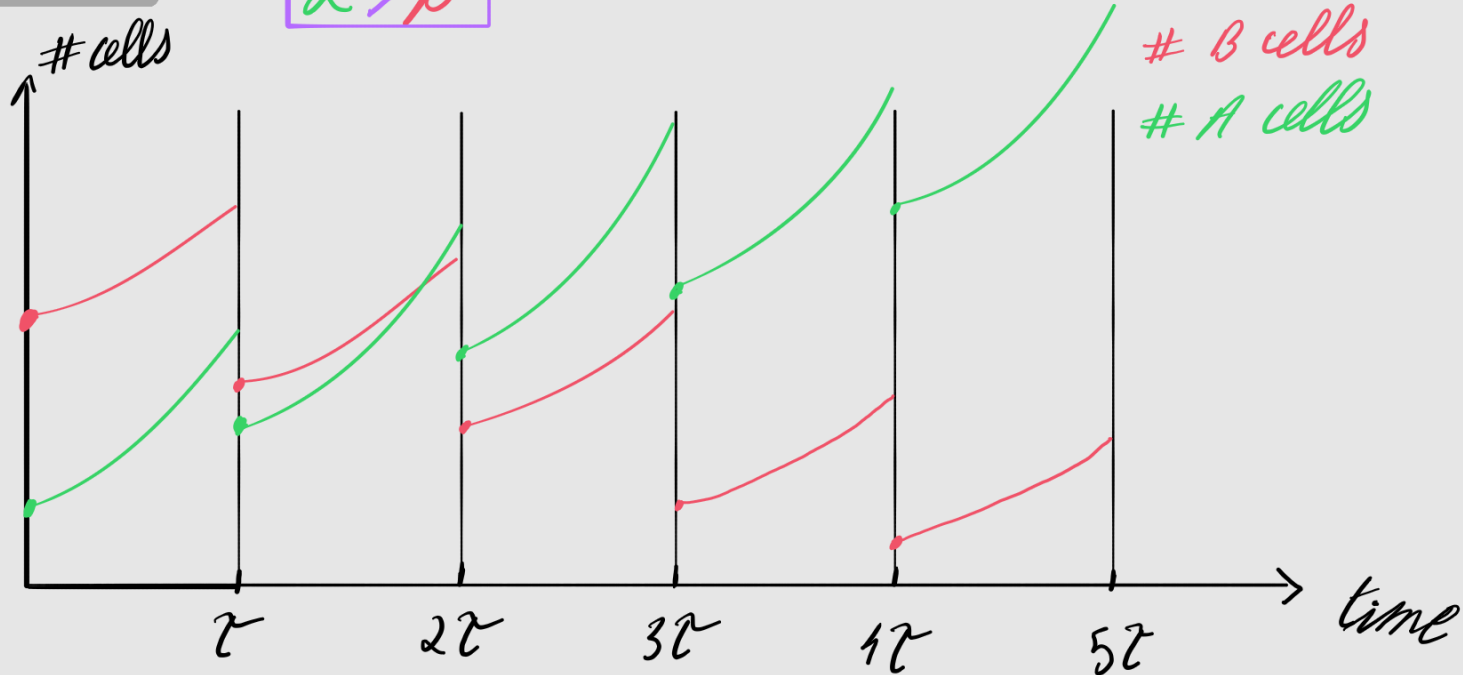


Problem 1: Population dynamics

1.a)

$$\alpha > \beta$$



1.b) Def: "Exponential growth" := any process in which the rate of change of a quantity, x , is proportional to its current value.

$x(t)$ exhibits exponential growth if it satisfies:

$$\frac{dx(t)}{dt} = \underbrace{k}_{\text{"growth rate"}} x(t), \quad x(t) = \text{size at time } t$$

for $k > 0 \Rightarrow$ the larger $x(t)$ gets, the faster it grows

$\frac{1}{x(t)} dx(t) = k dt$, integrating both sides:

$$\int_{x(0)}^{x(\tau)} \frac{1}{x} dx = \int_0^{\tau} k dt$$

$$\hookrightarrow \underbrace{\ln(x)}_0^{\tau} = k\tau \Rightarrow \boxed{x(\tau) = x(0) e^{k\tau}}$$

$$\ln(x(\tau)) - \ln(x(0)) = \ln\left(\frac{x(\tau)}{x(0)}\right)$$

Here, $x(0)$ represents the *initial quantity* at the start of the exponential growth phase.

Thus, after the i^{th} pipetting event, where we start in a fresh culture medium with, $N_{A,i}$ and $N_{B,i}$, after exponential growth for time τ (time before the next pipetting event), we have a total of cells equal to:

$$N(\# \text{ cells before pipetting event } i+1) = \underbrace{N_{A,i} e^{\alpha\tau}}_{\# A \text{ cells}} + \underbrace{N_{B,i} e^{\beta\tau}}_{\# B \text{ cells}}$$

1.c) $x(t) :=$ fraction of A cells

$1-x(t) :=$ fraction of B cells

$N :=$ total # of cells in the medium

Right after the i^{th} event:

$$x_i = \frac{n_{A,i}}{n_{A,i} + n_{B,i}}$$

$$N = n = n_{A,i} + n_{B,i}$$

1.d) Bacteria are technically "monoploid", they reproduce asexually and have 1 copy of each gene (each cell carries a single allele of type A or B)

Eq. (8) gives $D(x)$ for a haploid Wright-Fisher model,

using $N = n$:

$$D(x) = \frac{1}{2n} x(1-x)$$

⌚ In our experiment the randomness enters during the *pipetting* (dilution) step, which is a sampling process analogous to the Wright-Fisher model. Since bacteria reproduce asexually we used

the haploid version of the model.

1.e)

$$\text{Fraction just after event } i = x_i = \frac{n_{A,i}}{n}$$

$$\text{Fraction just before event } i+1 = x' = \frac{n_{A,i} e^{\alpha \tau}}{n_{A,i} e^{\alpha \tau} + n_{B,i} e^{\beta \tau}}$$

$$\text{using } n_{A,i} = n \cdot x_i, \quad n_{B,i} = n(1-x_i) = n - n_{A,i}$$

$$\text{we get: } x' = \frac{n \cdot x_i e^{\alpha \tau}}{n \cdot x_i e^{\alpha \tau} + n(1-x_i) e^{\beta \tau}}$$

$$\Rightarrow \Delta x' = x' - x_i = \frac{x_i e^{\alpha \tau}}{x_i e^{\alpha \tau} + (1-x_i) e^{\beta \tau}} - x_i$$

$$s = \alpha - \beta \Rightarrow \boxed{\Delta x' = \frac{x_i e^{s\tau}}{x_i e^{s\tau} + (1-x_i)} - x_i}$$

From eq.(2): $v(x) = \frac{\langle \Delta x' \rangle}{\Delta t}$, time between coarse-

grained steps is $\Delta t = \tau$, since the growth phase is

deterministic, $\langle \Delta x' \rangle = \Delta x'$. Replacing x_i with x :

$$\boxed{v(x) = \frac{1}{\tau} \Delta x' = \frac{1}{\tau} \left(\frac{x e^{s\tau}}{x e^{s\tau} + (1-x)} - x \right)}$$

For small $|(\alpha - \beta)\tau|$, we use $e^{s\tau} \approx 1 + s\tau$:

$$v(x) \simeq \frac{1}{\tau} \left[\frac{x(1+s\tau) - x^2(1+s\tau) - \cancel{x} + \cancel{x^2}}{x(1+s\tau) + (1-x)} \right]$$

$$= \frac{1}{\tau} \left[\frac{x(1-x)s\tau}{x \cdot s\tau + 1} \right]$$

Since $|s\tau|$ is small, $|s\tau x|$ is small and we use

$$\frac{1}{1+s\tau x} \simeq 1 - s\tau x$$

$$v(x) \simeq \frac{1}{\tau} (x(1-x)s\tau) (1 - s\tau x)$$

$$= \frac{1}{\tau} (x(1-x)s\tau + O(s^2\tau^2))$$

$$\Rightarrow \boxed{v(x) \simeq x(1-x)(\alpha-\beta)}, \text{ this is the drift velocity}$$

to 1st order in $\tau(\alpha-\beta) \equiv \tau s$, it matches the form of selection in a haploid population from eq. (5). $\alpha-\beta$, the difference in growth rates is the effective selection coefficient in our experiment.

1.f) We have $\begin{cases} v(x) \simeq x(1-x)(\alpha-\beta) & (c) \\ D(x) = \frac{1}{2n} x(1-x) & (d) \end{cases}$

Forward Kolmogorov:

$$\begin{aligned}\frac{\partial p(x,t)}{\partial t} &= -\frac{\partial}{\partial x} (v(x)p(x,t)) + \frac{\partial^2}{\partial x^2} [D(x)p(x,t)] \\ &= -\frac{\partial}{\partial x} (x(1-x)(2-\beta)p(x,t)) + \frac{1}{2n} \frac{\partial^2}{\partial x^2} (x(1-x)p(x,t))\end{aligned}$$

Steady-state $\Rightarrow \frac{\partial p^*(x,t)}{\partial t} = 0$

Assuming *no probability current* (like in the MIT notes):

$$J(x) = v(x)p^*(x) - \frac{d}{dx} (D(x)p^*(x)) = 0$$

$$\Leftrightarrow \frac{1}{D(x)p^*(x)} v(x)p^*(x) = \frac{1}{D(x)p^*(x)} \frac{d}{dx} (D(x)p^*(x))$$

$$\frac{d}{dx} \ln(D(x)p^*(x)) = \frac{v(x)}{D(x)}$$

$$\ln(D(x)p^*(x)) = \int^x \frac{v(x')}{D(x')} dx' + \text{const}$$

$$p^*(x) \propto \frac{1}{D(x)} e^{\int^x \frac{v(x')}{D(x')} dx'}$$

Plugging in (c) and (d) for $D(x)$ and $v(x)$:

$$\boxed{\rho^*(x) \propto \frac{1}{x^{1-2\alpha} (1-x)^{1-2\beta}} e^{\int_0^x \frac{\alpha-\beta}{x'} dx'}} \quad \boxed{\propto \frac{1}{x^{1-2\alpha} (1-x)^{1-2\beta}} e^{2n(\alpha-\beta)x}}$$

We have singularities at $x=0$ and $x=1$, thus there is *no stable steady-state distribution* where both A and B coexist ($0 < x < 1$). The non-normalizability on $[0,1]$ is a mathematical signal that the true long-term behavior is governed by absorbing states.

There are *no mutations* $\Rightarrow x=0$ & $x=1$ are the *absorbing states*. ($\int_0^1 \rho^*(x) dx$ diverges and is thus not a legitimate probability density, true steady state lives on absorption boundaries)

$\rightarrow$ The only steady states are fixation/loss, where the population consists entirely of one type. If there is a difference in growth rates ($s \neq 0$), selection determines the unique stable state ($x=1$ if $\alpha > \beta$ and $x=0$ if $\alpha < \beta$). If $\alpha = \beta$ both $x=1$ and $x=0$ are possible outcomes, at to 1st order $v(x) = 0$ ($\propto (\alpha - \beta)$), so the system reduces to pure genetic drift and the long-term behavior is *governed entirely by stochastic sampling during pipetting*.

Because there is no bias (no selection), the probability of eventual fixation of type A is simply equal to its initial frequency.

In the MIT notes we saw that for the "neutral" Wright-Fisher model the fixation probability is

$$\Pi_1(y) = y, \quad \Pi_0(y) = 1 - y. \quad (\text{eq. 1.73})$$

conclusion, in the long time limit ($t \rightarrow \infty$)

$\alpha > \beta$: $\rho^*(x) = \delta(x-1)$, A wins

$\alpha < \beta$: $\rho^*(x) = \delta(x)$, B wins

$\alpha = \beta$: $\rho^*(x) = y \delta(x-1) + (1-y) \delta(x)$, where

$$y = x_i = \frac{n_{A,i}}{n_{A,i} + n_{B,i}}$$

Problem 2: Population dynamics

$N = \text{const}$; the total population size is very large, $N \gg 1$

fraction of total alleles $x(t) = \frac{1}{1 + e^{-0.01t}}$

Not mutations, only inheritance

2.a) We have deterministic spread in a very large

population. Genetic drift is negligible: $N \gg 1 \Rightarrow D(x) = 0$

since $D(x) \propto \frac{1}{4N}$.

Humans are **diploid** $\Rightarrow v(x) = \frac{s}{2} x(1-x)$ (selection)

$\rightarrow$ Forward Kolmogorov: $\boxed{\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x} \left(\frac{s}{2} x(1-x) p(x,t) \right)}$

only the selection-driven velocity term is important

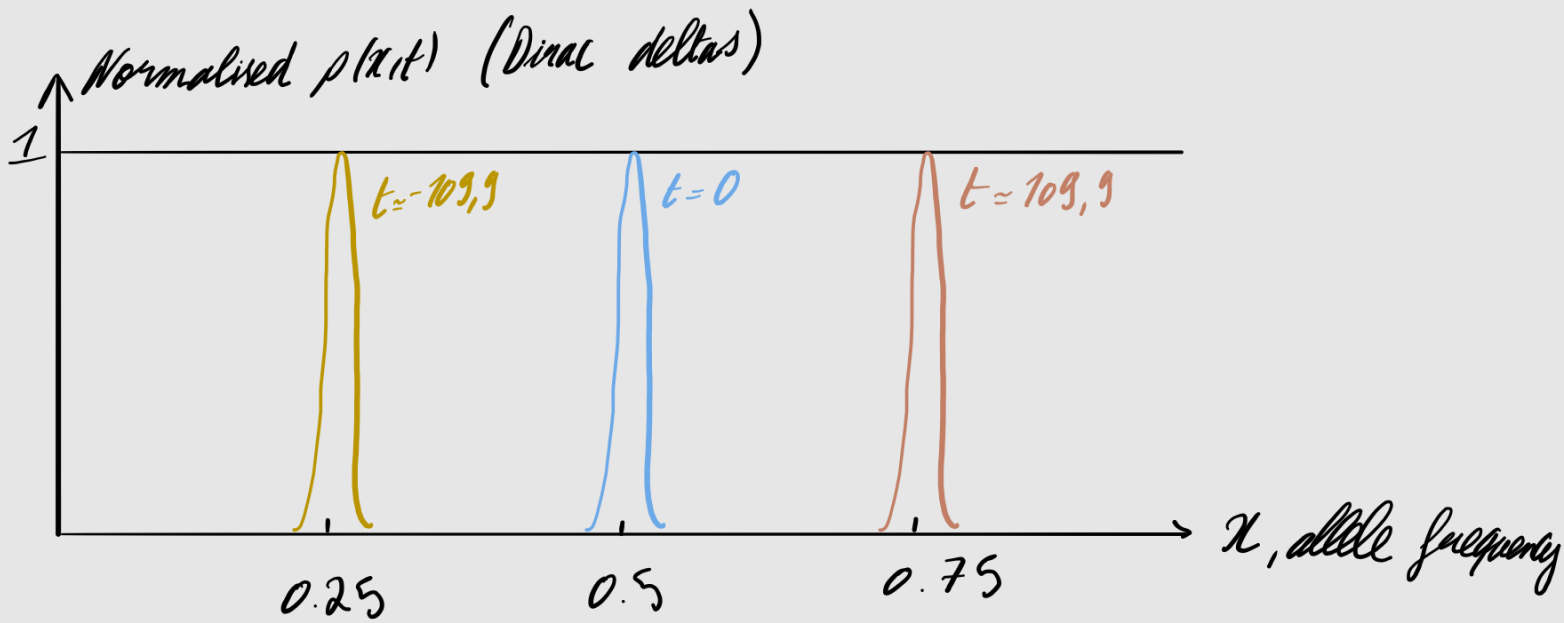
2. b) The only force acting is selection, which is deterministic, the allele frequency does not fluctuate, it follows the predictable path $x(t) = \frac{1}{1 + e^{-0.01t}}$

$p(x,t)$ is the probability density that the allele frequency is x at time t . In this deterministic system there is no uncertainty: at time t , the frequency must be $x(t)$. The distribution is thus a **Dirac delta function centered at $x(t)$** :

$$\boxed{p(x,t) = \delta(x - x(t))}$$

Over time, the delta function slides smoothly from $x=0$ to $x=1$, without broadening.

$$e^{-0.01t} = \frac{1}{x(t)} - 1 \Rightarrow t = -100 \ln\left(\frac{1}{x(t)} - 1\right)$$



2.C) Following the hint:

$$\int_0^x \frac{\partial p(x',t)}{\partial t} dx' = \int_0^x -\frac{\partial}{\partial x'} \left(\frac{s}{2} x'(1-x') p(x',t) \right) dx'$$

$$\frac{\partial c(x,t)}{\partial t} = -\frac{s}{2} x(1-x) p(x,t) = -\frac{s}{2} x(1-x) \frac{\partial c(x,t)}{\partial x}$$

where we assumed the swap of integration and ∂_t is

valid and used $c(x,t) = \int_0^x p(x',t) dx' \Rightarrow p(x,t) = \frac{\partial c(x,t)}{\partial x}$

We use the ansatz $c(x,t) = F(z)$ where $z = f(x) + g(t)$

$$\hookrightarrow \frac{\partial F}{\partial z} \frac{\partial g}{\partial t} = -\frac{s}{2} x(1-x) \frac{\partial F}{\partial z} \frac{\partial f}{\partial x}$$

assuming $\partial_z F \neq 0$: $g'(t) = -\frac{s}{2} x(1-x) f'(x)$

LHS is a function of t only, RHS is a function of x only,

for the equality to hold for all x & t , both sides must be equal to the same constant K

$$\begin{cases} g'(t) = K \Rightarrow \boxed{g(t) = Kt + C_1}, \quad C_1 = \text{const} \\ -\frac{5}{2}x(1-x)f'(x) = K \Rightarrow f'(x) = \underbrace{\frac{-2K}{5}}_{=K'} \frac{1}{x(1-x)} \end{cases}$$

$$f'(x) = K' \cdot \frac{1}{x(1-x)} = K' \left(\frac{1}{x} + \frac{1}{1-x} \right)$$

$$\hookrightarrow f(x) = K' [\ln|x| - \ln|1-x|] + C_2$$

$$\boxed{f(x) = -\frac{2K}{5} \ln \left| \frac{x}{1-x} \right| + C_2}, \quad C_2 \text{ const}$$

$$z = f(x) + g(x) = -\frac{2K}{5} \ln \left| \frac{x}{1-x} \right| + Kt + (C_1 + C_2)$$

We can absorb constants into the definition of F .

We can choose a convenient $K = -\frac{5}{2}$, giving us:

$$\boxed{C(x,t) = F\left(\underbrace{\ln\left(\frac{x}{1-x}\right) - \frac{5}{2}t}_{z'}\right)} \quad (0 < x < 1)$$

We know $\rho(x, 0) = \delta(x - 0.5)$, and $\delta(x) = \frac{d}{dx} \Theta(x)$ where Θ is the Heaviside step function. Therefore

$$\delta(x - 0.5) = \rho(x, 0) = \frac{\partial}{\partial x} \Theta(x - 0.5) = \frac{\partial c(x, 0)}{\partial x}$$

$$\text{so } c(x, 0) = \Theta(x - 0.5) = F\left(\ln\left(\frac{x}{1-x}\right)\right)$$

Let $u = \ln\left(\frac{x}{1-x}\right)$: for $x = 0.5$, $u = 0$

if $x < 0.5$, then $u < 0$, $F(u) = \Theta(x - 0.5) = 0$

if $x > 0.5$, then $u > 0$, $F(u) = \Theta(x - 0.5) = 1$

Notice that $F(u) = \begin{cases} 0 & \text{if } u < 0 \\ 1 & \text{if } u > 0 \end{cases}$ is exactly

the definition of the Heaviside step function.

Therefore $F(u) = \Theta(u)$, giving us $c(x, t)$:

$$\Rightarrow c(x, t) = F(z') = \Theta(z') = \Theta\left(\ln\left(\frac{x}{1-x}\right) - \frac{s}{2}t\right)$$

Final form of the probability density

$$p(x,t) = \frac{\partial C(x,t)}{\partial x} = \delta\left(\ln\left(\frac{x}{1-x}\right) - \frac{s}{2}t\right) \underbrace{\frac{d}{dx}\left[\ln\frac{x}{1-x}\right]}_{\frac{1-x}{x} \cdot \left(\frac{1}{1-x} + \frac{x}{(1-x)^2}\right) = \frac{1}{x(1-x)}}$$

Using $\delta(h(x)) = \frac{\delta(x-x_0)}{|h'(x_0)|}$ where $h(x_0) = 0$ with

$$h(x) = \ln\left(\frac{x}{1-x}\right) - \frac{s}{2}t \Rightarrow h'(x) = \frac{1}{x(1-x)}$$

$$h(x_0) = 0 \Rightarrow \ln\left(\frac{x_0}{1-x_0}\right) = \frac{s}{2}t \Rightarrow x_0 = (1-x_0)e^{\frac{s}{2}t} \\ \Rightarrow x_0 = \frac{e^{\frac{s}{2}t}}{1 + e^{\frac{s}{2}t}}$$

$$\Rightarrow x_0 = \frac{1}{1 + e^{-\frac{s}{2}t}}$$

$|h'(x_0)|$ cancel with the $\frac{d}{dx}\left[\ln\left(\frac{x}{1-x}\right)\right]$ term (we have a

Dirac delta so $x = x_0$ is our only concern), leaving us with:

$$p(x,t) = \delta(x-x_0) = \delta\left(x - \frac{1}{1 + e^{-\frac{s}{2}t}}\right)$$

2.d) Comparing the above equation with 2.c)

we get : $\frac{s}{2} = 0.01 \implies \boxed{s = 0.02}$

this coefficient determines the observed spread of the pink-eye allele, representing a 2% fitness advantage in the diploid human population.

2.e) Diploid population with N individuals has $2N$ alleles.

Amongst the 10 people we initially have 1 allele for pink-eyes $\implies$ Initial fraction: $y = \frac{1}{20}$

from eq. 10 : probability of fixation for an allele with initial fraction y is: $\Pi_1(y) = \frac{1 - e^{-2sNy}}{1 - e^{-2sN}}$

with $s = 0.02$, $N = 10$, $y = \frac{1}{20} \implies \Pi_1(y) \approx 0.0601$

i.e. $\boxed{\approx 6\%}$ chance of fixation